

Nash Equilibrium: Existence I

Let $G = (N, \{S_i\}_{i \in N}, \{u_i\}_{i \in N})$ be a strategic form game with $N = \{1, \dots, n\}$ for some positive integer n .

Theorem (Debreu-Fan-Glicksberg)¹ G has a Nash Equilibrium if, for every $i \in N$,

- (a) S_i is a nonempty, compact, convex subset of some Euclidean space;
- (b) u_i is quasiconcave on S_i for every $s_{-i} \in S_{-i}$; and is continuous in (s_i, s_{-i}) on S .

The underlined assumptions ensure that best replies are nonempty. Now suppose that G is a finite game (each S_i is finite), and denote its mixed extension by $\Delta G = (N, \{\Delta S_i\}_{i \in N}, \{u_i\}_{i \in N})$, where for each $i \in N$, ΔS_i is the set of probability distributions over S_i , and u_i is a vN-M utility (what MWG call a Bernoulli utility). I denote a typical element of ΔS_i by σ_i ; the probability that player i plays ‘pure’ strategy $\hat{s}_i \in S_i$ by $\sigma_i(\hat{s}_i)$; a mixed strategy profile by $\sigma = (\sigma_1, \dots, \sigma_n)$, where $\sigma_i \in \Delta S_i$ for all $i \in N$; and the set of feasible strategy profiles for ΔG by ΔS . Note that the expected utility for player i when profile σ is played is

$$U_i(\sigma) = \sum_{s \in S} u_i(s) \sigma_1(s_1) \sigma_2(s_2) \cdots \sigma_n(s_n).$$

[The probability that pure strategy $s = (s_1, s_2, \dots, s_n)$ is played = (the probability that player 1 plays s_1) $\times$ (the probability that player 2 plays s_2) $\times \cdots \times$ (the probability that player n plays s_n).]

Corollary (Nash, 1950) A mixed extension ΔG of a finite game G has a Nash equilibrium.

Fact For a finite game G , $\sigma^* \in \Delta S$ is a Nash equilibrium of ΔG if and only if for every $i \in N$, $\sigma_i^*(\hat{s}_i) > 0$ implies that $\hat{s}_i$ is a best reply to σ_{-i}^* . In words, iff any ‘pure’ strategy that player i plays with positive probability is itself a best reply to the mixed strategy profile of the other players. In particular, each player i must be indifferent between any two of its pure strategies played with positive probability, and any of these must be at least as good against σ_{-i}^* as any strategy played with zero probability.²

Example: Consider a two-player game G with $S_1 = S_2 = [a, b] \subset \mathbb{R}$, $a < b$, and each u_i strictly quasiconcave in s_i and continuous in s . Let $r_i : [a, b] \rightarrow [a, b]$ be player i ’s best reply mapping. Since u_i is continuous in s_i on the compact set $[a, b]$, r_i is nonempty; and since u_i is strictly quasiconcave, it is single-valued. Define the function $g : [a, b] \rightarrow [a, b]$ by $g(s_1) = r_1(r_2(s_1))$. By the Theorem of the Maximum, each r_i is continuous, so g is continuous. Since in addition $g(a) \geq a$ and $g(b) \leq b$, it follows from the Intermediate Value Theorem that there is a point $s_1^* \in [a, b]$ with $g(s_1^*) = s_1^*$. The point $(s_1^*, r_2(s_1^*))$ is a Nash equilibrium of G .

Proof of Theorem: Let $r_i : S_{-i} \rightarrow 2^{S_i}$ be player i ’s best-reply correspondence. Under (a) and (b), it is nonempty and upper semicontinuous by the Maximum Theorem, and convex-valued since u_i is quasiconcave in s_i . Define the correspondence $r : S \rightarrow 2^S$ by $r(s) = r_1(s_{-1}) \times \dots \times r_n(s_{-n})$, which is also nonempty, upper semicontinuous and convex-valued. Since S is nonempty, convex, and compact, by Kakutani’s Fixed Point Theorem there is a point $s \in r(s)$. (Kakutani’s Theorem: if Y is a nonempty, convex, compact subset of a Euclidean space and $\phi : Y \rightarrow 2^Y$ is an upper semicontinuous, convex-valued correspondence, then ϕ has a fixed point. See MWG, p. 953.). ■

¹Debreu, G., 1952. “A social equilibrium existence theorem” *Proceedings of the National Academy of Sciences* 38, 886-893. Fan, K., 1952. Fixed point and minimax theorems in locally convex topological linear spaces. *Proceedings of the National Academy of Science* 38, 121-126. Glicksberg, I., 1952. A further generalization of the Kakutani fixed point theorem with application to Nash equilibrium points. *Proceedings of the American Math Society* 3, 170-174.

²I give a proof of this Fact on page 2.

A NOTE ON AFFINE PROGRAMMING

Let $P = \{p \in \mathbb{R}_+^m \mid \sum_j p_j = 1\}$ and define

$$P^* = \arg \max_{p \in P} p \cdot y$$

where $y \in \mathbb{R}^m$. This is an affine program because $p \cdot y$ is both convex and concave in the choice, p .

Fact 0 These two things are equivalent.

1. $p^* \in P^*$
2. $p_j^* > 0$ implies that $e_j \in P^*$.

(As before e_j is a vector of 0's with a 1 in the j th place.)

Proof of Fact 0: Let 2. hold. Since P^* is convex, if $e_j \in P^*$ for every j with $p_j^* > 0$, then $p^* = \sum_j p_j^* e_j \in P^*$, so 1. holds. Now suppose that 2. fails, that is $p_j^* > 0$ for some j with $e_j \notin P^*$. Then there is a $1 \leq k \leq m$, $k \neq j$, with $y_k > y_j$. Define p' by $p'_j = 0$ and $p'_k = p_k^* + p_j^*$, with all other coordinates equal: $p'_i = p_i^*$ for $k \neq i \neq j$. Then $p' \in P$ and $p' \cdot y - p^* \cdot y = p_j^*(y_k - y_j) > 0$, so 1. fails. It follows that 1. implies 2. $\square$

In words: p^* solves the affine optimization problem if and only if, for each j with $p_j^* > 0$, e_j solves the problem. Now apply Fact 0 to this Fact, taken from my notes on Nash Equilibrium: Existence I.

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Consider player i . Let m be the number of player i ’s strategies in S_i , so we can write $S_i = \{s_{i1}, \dots, s_{im}\}$. The set P equals ΔS_i , the strategy set of i in the mixed extension of G , ΔG . Set

$$y_j = \sum_{s_{-i} \in S_{-i}} u_i(s_{ij}, s_{-i}) [\prod_{k \neq i} \sigma_k(s_k)]^3$$

Then $p \cdot y$ is player i ’s expected utility when i uses strategy $p \in \Delta S_i$.

Applied to the matching pennies examples, it is easy to conjecture and verify that a Nash equilibrium of the mixed extension is that each player chooses H with probability 1/2. Of course, if player j chooses H with probability 1/2, then player $i \neq j$ ’s best response is to choose H for sure, L for sure, *or any probability mixture of those two*, including 1/2 each. This fact will be useful to you in finding all equilibria in the mixed extension of the Battle of the Sexes game.

³ $\prod_{k \neq i} \sigma_k(s_k)$ is simply the probability of s_{-i}